

Script of the Design Course

Design Course: RF- and Microwave Circuits and Systems

Prof. Dr. Rüdiger Quay,

ruediger.quay@inatech.uni-freiburg.de

19.04.2026

Version 12.0



EEH, Fritz-Hüttinger chair energy-efficient
high-frequency electronics
Freiburg University

Contents

1	Abstract	6
2	Elements of PDK: Transistor Models	9
2.1	Active Modeling: Small-Signal Approach	9
2.2	Task 6 and 7: Small-Signal Models (Block 2A)	15

Introduction

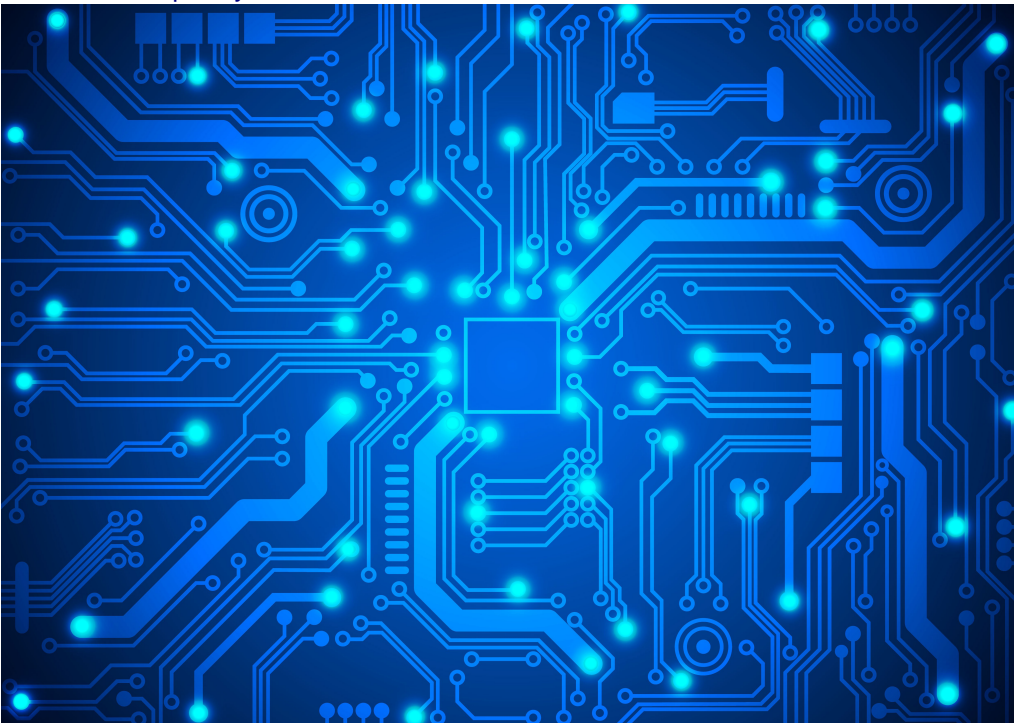


Introduction



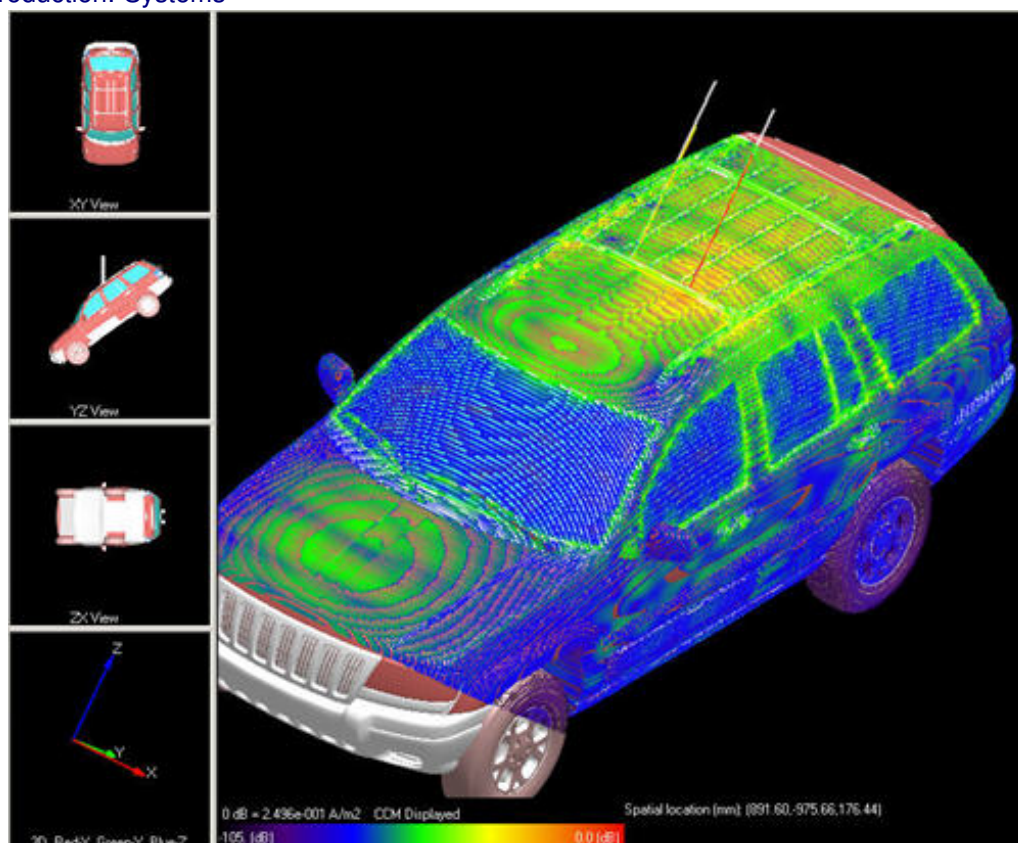
1.4

Introduction: Complexity



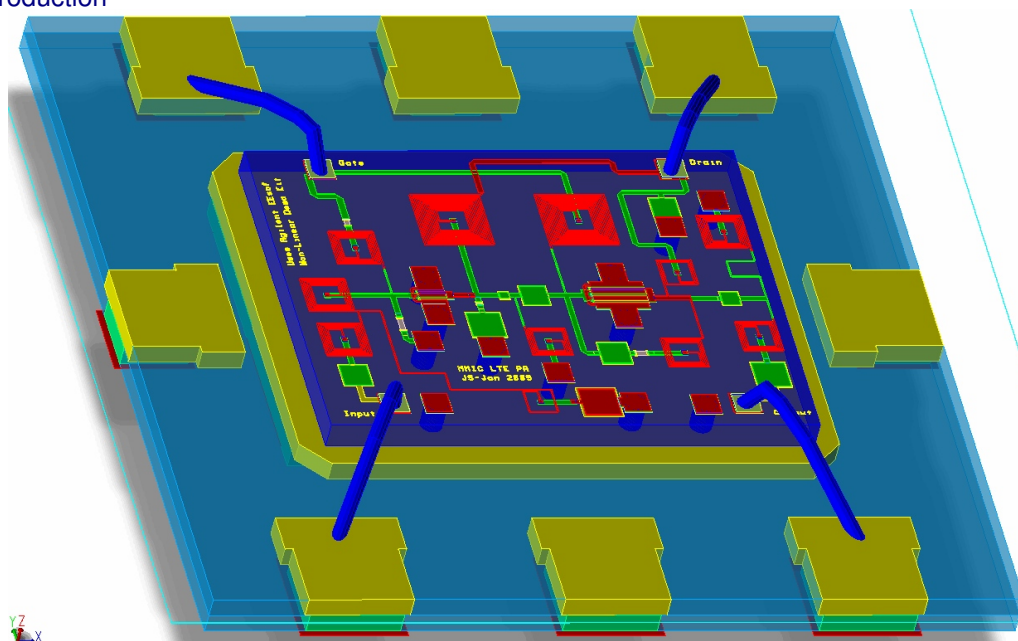
1.5

Introduction: Systems



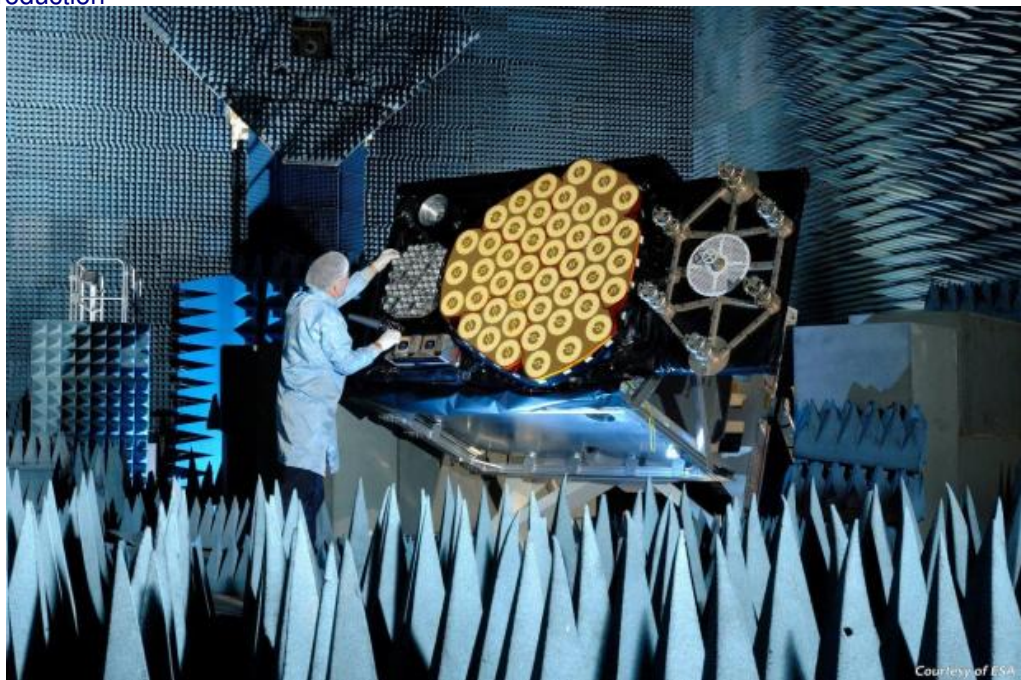
1.6

Introduction



!!!! Frequency domain !!!! vs. Cadence Virtuoso: Time domain

1.7



Dates with exercise

- Mo 1.) 20.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 2.) 27.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 3.) 4.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 4.) 11.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 5. 18.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 6. 25.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 7. 1.06.2026 no lecture, pentecost break SR 00 014 (G.-Köhler-Allee 078)

1.9

Dates with exercise II

- Mo 8. 8.6.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 9. 15.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 10. 22.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 11. 29.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 6.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 13.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 13. 20.07.2026 Presentation of final results from competition 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)

1.10

Date and Location

Dates with exercise

1. Mo, 20.4.2026: 16-17.00 h Introduction and group formation
2. Block 1: Mo., 27.4.2026: (issue block 1a on ILIAS)
3. Block 1: Mo., 4.5.2026: (issue block 1b on ILIAS)
4. Block 2: Mo., 11.5.2026 (issue block 2a on ILIAS), (Deadl. Block 1)
5. Block 2: Mo., 18.5.2026 (issue block 2b on ILIAS)
6. Block 3: Mo., 25.5.2026 (deadline block 2 on ILIAS), (issue block 3a on ILIAS)
7. Block 3: Mo., 08.6.2026 (issue block 3b on ILIAS)
8. Block 4: Mo., 15.6.2026 (Deadline block 3), (issue block 4a on ILIAS)
9. Block 4: Mo., 22.6.2026, (Issue Block 4b, COMPETITION !!!)
10. Block 5: Mo. 29.6.2026 (deadline block 4), (issue block 5a on ILIAS)
11. Block 5: Mo., 06.7.2026 : (issue block 5b ON ILIAS)
12. Mo., 13.7.2026, Final presentations based on 4b (deadline block 5 on ILIAS)

1.11

Rules of the Exercises

- 5 blocks Exercises (every one/two weeks) with overall max. 10 tasks in a given format...
- First: starting 27.4.2026... (two weeks to do, sometimes more)
- 4 blocks to be completed successfully (with 50% correct) (three people per group)
- Voluntary Competition: Final presentation (5 mins) of the group to the plenary based on a task
- Exercises are given out on ILIAS and also submitted there.
- **Timing is day of the deadline, 16 h**

1.12

1 Abstract

The students will be enabled to understand, design and layout modern RF- and microwave components and systems by means of the electronic design environment Advanced Design System including the two- and three dimensional electromagnetic simulators Momentum and EMPro 3D. The detailed use of a complex RF-software environment is a dedicated target of this course. This includes the numerical analysis of complex passive and active devices, the design and layout of hybrid and integrated circuits, and their packaging and signal flow. The students will be competent to design and layout passive and active RF-structures including packages and interconnects and circuits of relevance to everyday communication and sensing. The competence includes in-depth understanding and treatment of complex microwave systems and of general system design including the treatment of complex modulated signal flows.

Introduction

Materials and Script

The material is provided at the beginning and during the lecture.

Email: ruediger.quay@inatech.uni-freiburg.de

- Foils: Intro materials from Keysight
- ADS Video Library on www.youtube.com
- Script (supporting materials)
- Exercises

1.13

Videos

1. Watch ADS videos on youtube using ADS : www.youtube.com , Keyword: ADS Video Library
2. Basics and setup:
3. <https://www.youtube.com/watch?v=04BfajzXnmM&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz>
4. Understanding Workspaces:
5. <https://www.youtube.com/watch?v=7jNm400BaiE&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=2>
6. The basics of schematics:
7. <https://www.youtube.com/watch?v=wqyyOCzxGS0&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=4>
8. Data displays:
9. <https://www.youtube.com/watch?v=jIg10RihLQk&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=6>

1.14

WWW Support 1

ILIAS Support

<https://ilias.uni-freiburg.de>

search for: Design Course: RF- and Microwave Systems

password: design2026

1.15

WWW Support 2

This exercise is recorded, attendance is, however, required



1.16

Windows Access to the Uni Network

- Get login at university, get additional TF login
- Install VPN from University server
- use ssh from linux
- If you start from Windows:
- Use, e.g., xming as windows X-server (for export of the display)
- Install Putty ssh client on windows

1.17

Form groups of Three

- The exercises are submitted from a group of three people
- Find two partners !!!
- Send your group to ruediger.quay@inatech.uni-freiburg.de, including TF login name.
- Exercises start april 27th 2026 !

1.18

Install an ADS Version on a local computer

Last Updated: 05 March 2020

ADS - Windows® Operating Systems	
Product	Windows
ADS 2020 / ADS 2020 Update 1.0 / ADS 2020 Update 2.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2019/ ADS 2019 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2017/ ADS 2017 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2016.01 64-bit Operating Systems	Windows 10 (64-bit)* Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2015.01 64-bit Operating Systems	Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)

You can work on the aragorn but local has proven to be easier...

Organize your data in a good structure

1.19

Login to Application Server for the license

From Remote: Create RAS Access in Mylogin

Go into University net with VPN: check VPN Uni Freiburg

In case you use Windows to do that: use putty and install xming as windows X-server

Application Server: `aragorn.imtek.uni-freiburg.de`, `ssh-X-1 (yourlogin) aragorn.imtek.uni-freiburg.de`

Shells usable: `bash` and `tcsh`

Start: `ads` after using `ssh` to the server

Licence Server will be asked: `27010(at)frodo.imtek.uni-freiburg.de`

1.20

Backup Server !

Application Server: `samweis.imtek.uni-freiburg.de` (only via VPN)

1.21

Recommendation

Use Licence server only, install ADS on your local device ! Use ADS 2023 Update 1.0

1.22

The following references may be useful to provide more detailed information.

References

- List of useful ADS Videos
- Keysight Software Support: <http://www.keysight.com/>
- Introduction to ADS, ADS_Basics.pdf (ILIAS)
- Momentum Momentum_Basics.pdf (ILIAS)

1.23

How to simulate and understand the active device...

12. may 2026

3.24

2 Elements of PDK: Transistor Models

2.1 Active Modeling: Small-Signal Approach

Educational Targets for Block 2

- Understand how models are (re)presented in ADS (open, just S-parameters, bias-dependent models, just data).
- Play with RF-parameters in ADS (S-parameters, gains such U,...)
- Understand active DC models...
- Understand small-signal models for the description of transistors in the frequency domain.
- Understand the composition of an active small-signal models for field effect transistors.
- Understand the meaning and impact of passive parasitic elements in active devices.
- Understand the active aspect in a small-signal representation.
- Understand first ideas of large-signal approaches.

3.25

Educational Targets

Various small-signal models:

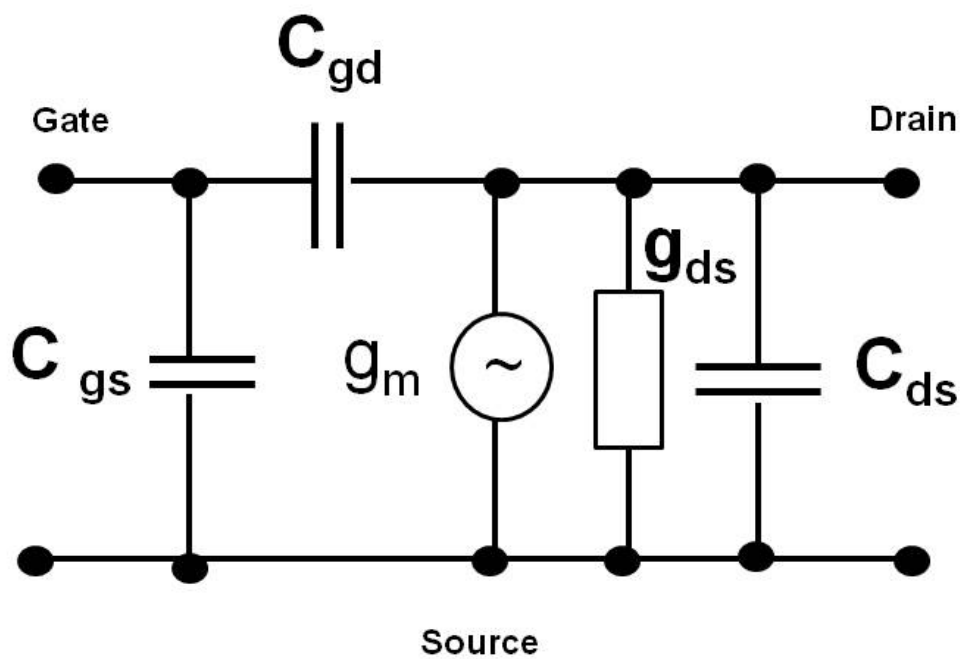
- Models can be: Just data in a data set
- Bias-dependent DC and RF-data (S-parameters) given as data sets....
- Equations valid for one bias point and for set of parameters (current, voltage)
- Bias-dependent equations for models (Large signal idea)

3.26

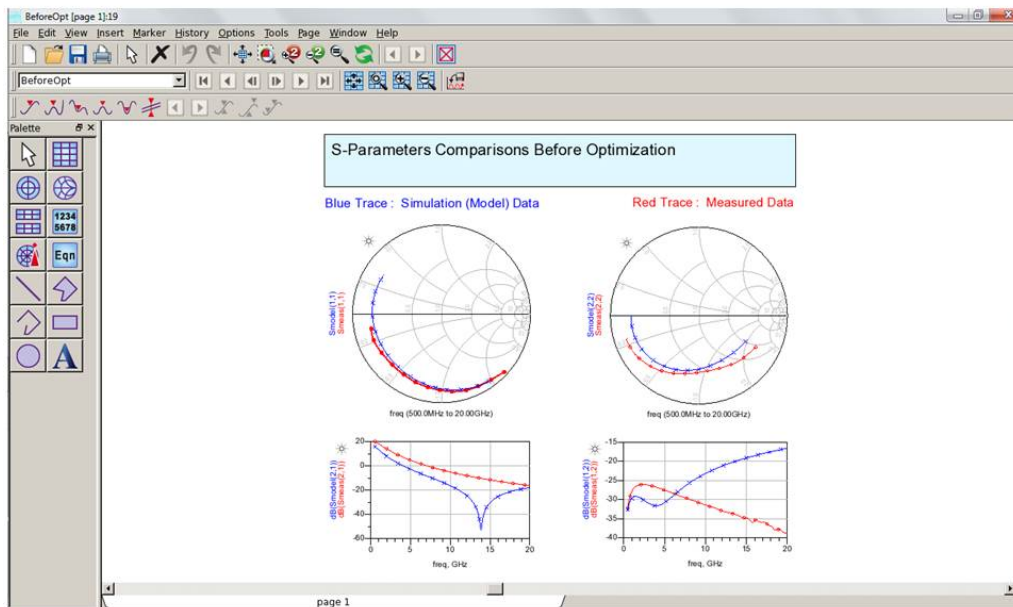
Videos for Active Interactions: how are things done ?

- DC-simulation setup <https://www.youtube.com/watch?v=SU4qpcSsXLA>
- AC-simulation setup https://www.youtube.com/watch?v=RzI_vz4Yui4
- Modeling ideas: Deembedding: https://www.youtube.com/watch?v=0BZSXz02y7o&list=PLtq84kH8xZ9H_6rL6xajTNW6_lmQMucR2&index=2
- Modeling of a bipolar transistor (advanced, just to see the procedure) <https://www.youtube.com/watch?v=UEqf0vDhlcw>

3.27

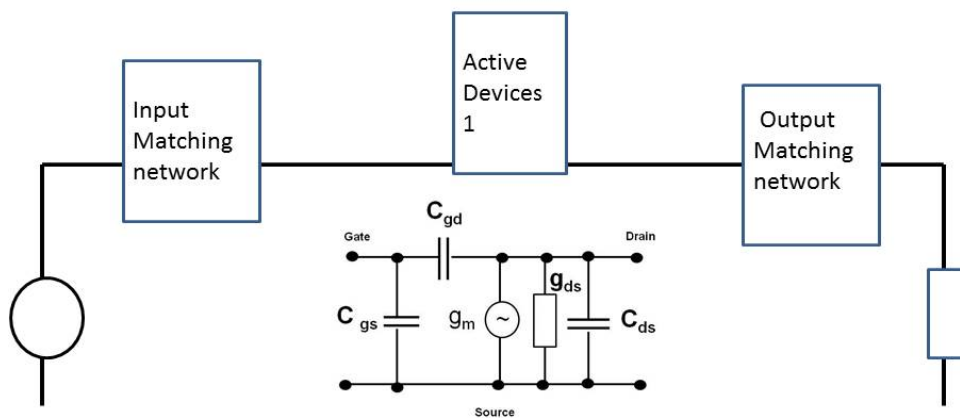


Equivalent Circuit: Circuit in ADS

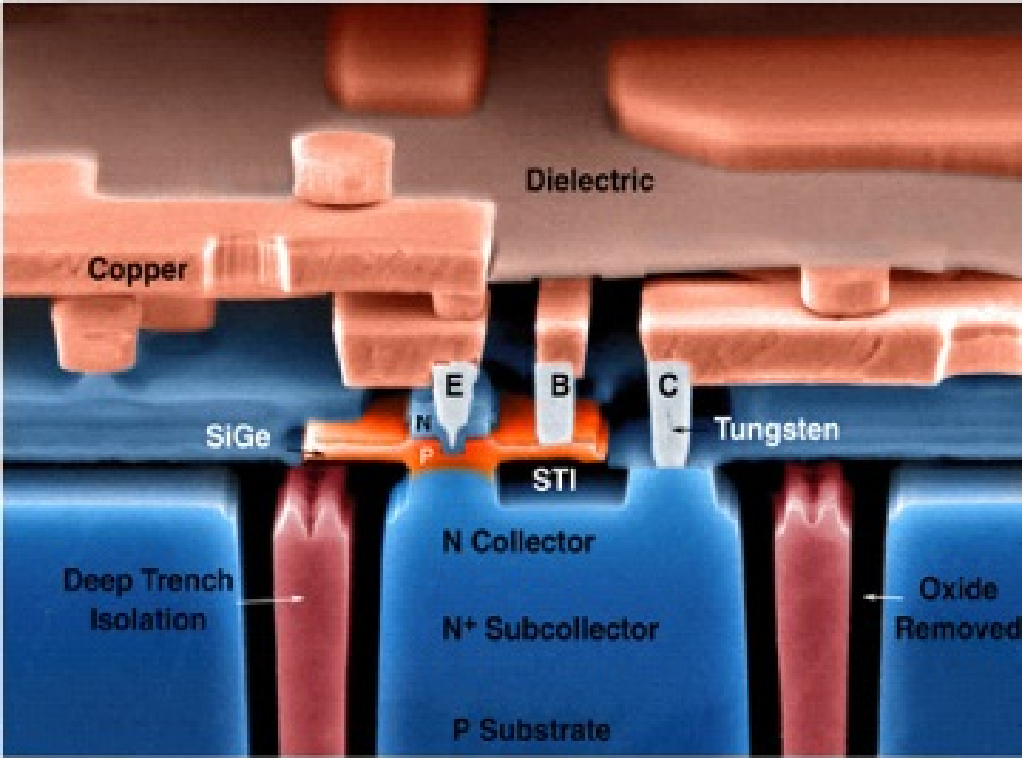


3.29

Equivalent Circuit FET: Parasitic Structure and Deembedding



3.30



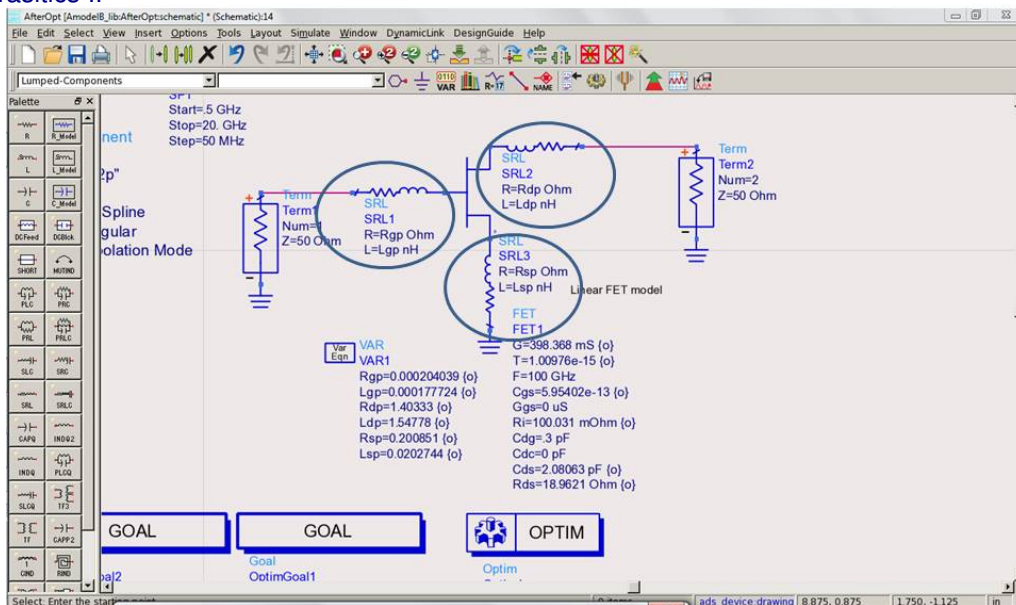
Equivalent Circuit: Small-signal Model in ADS

Example: FET2 Model with Source Resistance

A screenshot of the ADS 2012 Help window. The window title is 'ADS 2012 Help'. The address bar shows a path to a help file. The left pane shows a 'Contents' list with various models, including 'FET2 (Field Effect Transistor with Source Resistance)' which is highlighted. The right pane shows the details for 'FET2 (Field Effect Transistor with Source Resistance)'. It includes a 'Symbol' section with a diagram of a transistor with three terminals labeled 1, 2, and 3. Below the symbol is a 'Parameters' section with a table listing parameters: G (magnitude of transconductance at DC), T (time delay associated with transconductance), F (transconductance roll-off frequency), and Cgs (gate-to-source capacitance).

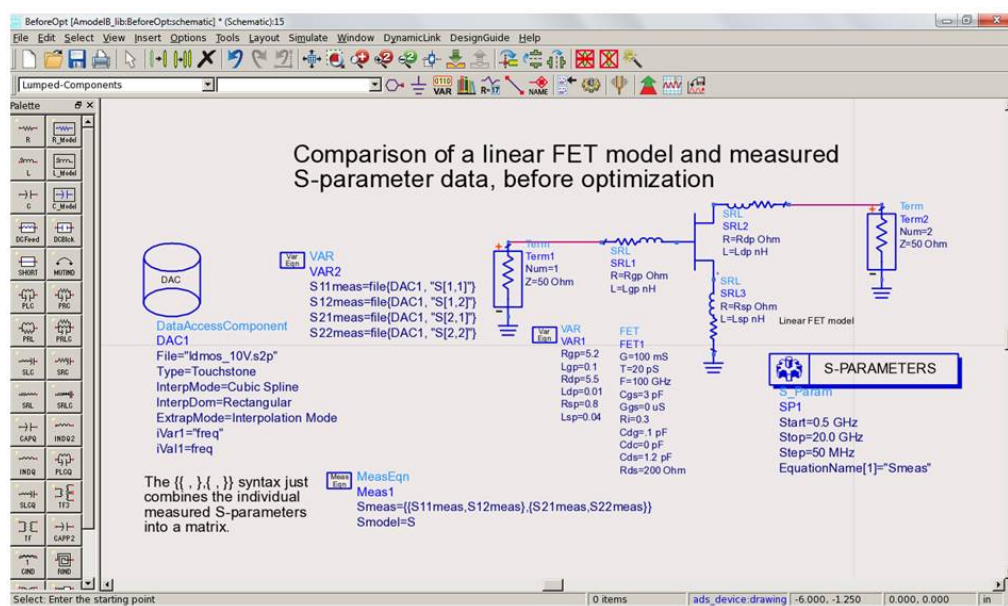
Name	Description	Units	Default
G	magnitude of transconductance at DC	uS	20.0
T	time delay associated with transconductance	nsec	1.0
F	transconductance roll-off frequency	GHz	1.0
Cgs	gate-to-source capacitance	pF	10.0

Parasitics II



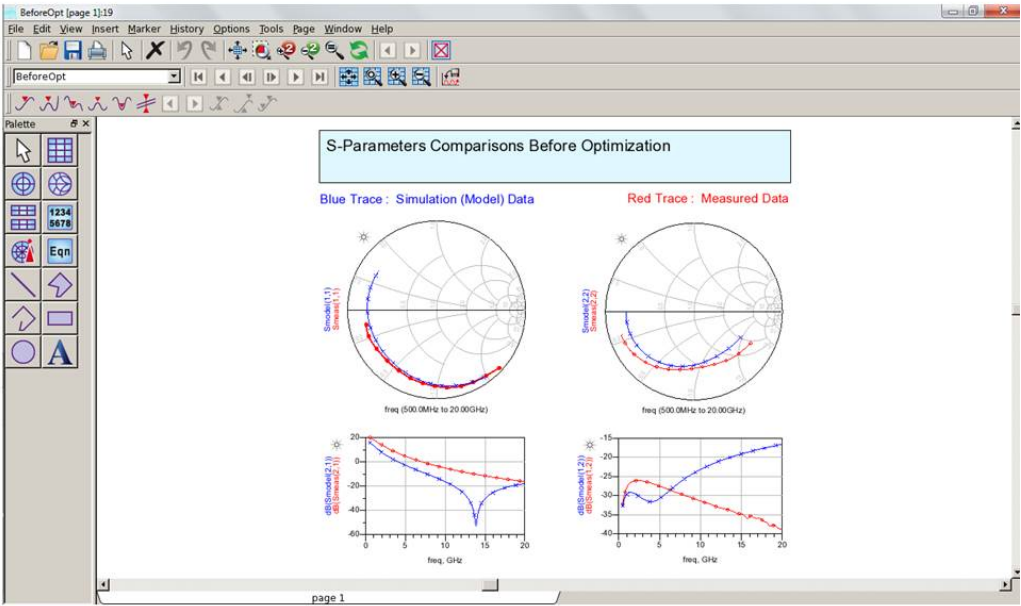
3.33

Equivalent Circuit: Circuit in ADS



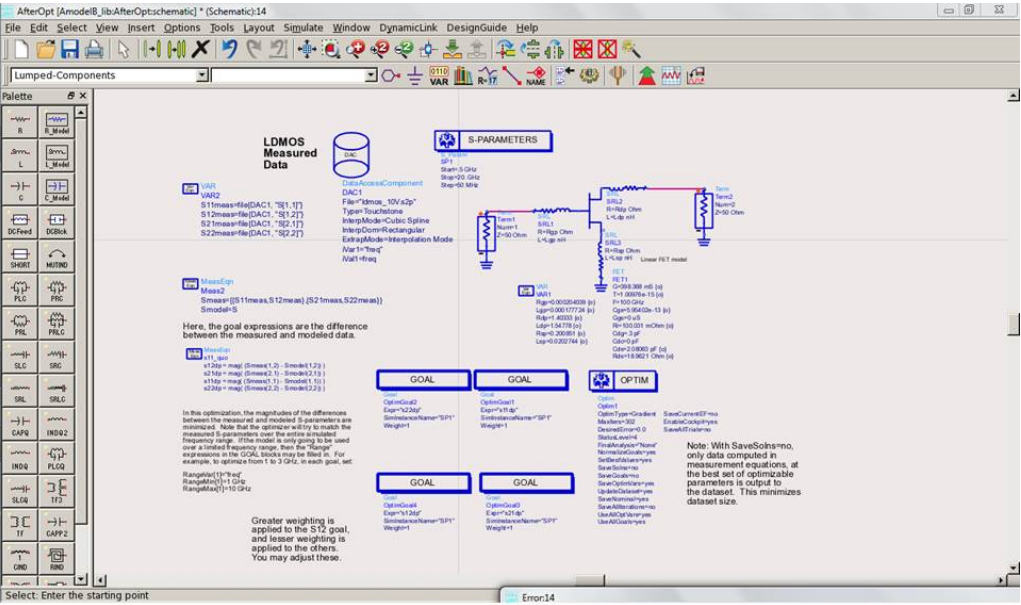
3.34

Equivalent Circuit: Circuit in ADS



3.35

Optimizer



3.36

2.2 Task 6 and 7: Small-Signal Models (Block 2A)

Tasks 6 and 7: Small Signal Models of a FET and a Bipolar Junction Transistor (BJT) (Block 2A)

1. Explanation: Go to help, read the text in help/, search for: /Examples of S-parameter simulation.
2. Explanation: Take the example `AmodelB_wrk` from the examples/MW circuits. This is an optimization example: it will help you to understand how models are generated from measurements, and what is a good and a poor model. The answer, as always, is: it depends.
3. Explanation: Look at the BeforeOpt dds (Data display). A model is made to describe the measured S-parameters (red) by resimulating a network (blue). Click on the model and check what it is. The parameter entry mode is standard, i.e., it is small-signal equivalent element-based.
Remember or learn: The S-parameters give you reflection coefficients based on a reference impedance of 50 Ohms in magnitude and phase.
4. Task: Give the impedances represented by S11 and S22 at 2.5 GHz for the model (blue) and the measured data (red). Compare the discrepancy in real and imaginary IMPEDANCE as a delta (i.e. in Ohms)! 5 points

3.37

Tasks 6 and 7: Small Signal Models of a FET and a Bipolar Junction Transistor (BJT) (Block 2A)

5. Compare the deviation of measurements and simulation.
In order to assess what is wrong or better unaccounted with this model:
6. Task: In order to do this: a.) look at the lowest frequency range for S11 and S22 (where is that in the Smith/Polar Chart?), which deviations (give the delta in magnitude/phase) do you see ? (2 sentences) 5 points
7. Look at a high frequency (i.e. 14 GHz): Again calculate the magnitude and the phase of the impedance for both cases (S11, S22) and compare. (2 Sentences) 5 points
8. Task: Looking at the S12 and its magnitude: what is the value in absolute numbers at 14 GHz ? What is the ratio of $|S21|/|S12|$? (3 sentences) 5 points
9. The next section is devoted to how to improve a model: Task: Go to schematic After opt. Press the optimizer button (red/green arrow). Plot the results you get after 25 steps (just the plot) 5 points

3.38

Model continued

10. Task: Have the optimizer running for a minimum of 150 steps and observe the movement of the optimization.
11. Task: Explain in 5 sentences max. what an optimizer does and why there can be still deviations after optimization (5 points)
12. Task: To get better results: Reduce the frequency range, in which you optimize from 0.5 GHz to 2 GHz only. After you completed the new optimization, resimulate the S-parameters and plot them. (5 points)
13. Task: Check again S12 optimization vs. measurements (Plot Mag and phase)... Up to which frequency do you consider it o.k. in magnitude and phase ? (5 points)

3.39

Tasks 6 and 7: Small Signal Models of a Bipolar Junction Transistor (Block 2 Part 1)

14. Explanation: Again, the next bipolar junction (BJT) transistor example is a full one, however, do get confused by the high number of schematics provided.
15. Explanation: Open the project: `examples/MW_CKTS/LNA_wrk`
16. Task: Use the curve tracer function to plot the bipolar output curve current ICE (collector emitter output current) of the BJT between $V_{CE} = 0...3$ V and $I_B = 0...40$ μ A with 10 μ A Steps. Which effect does happen in a real bipolar for high V_{CE} and high current ? (2 sentences) (5 points)
17. Task: Go to schematic `DC_op_point` and plot the bipolar transfer curve ICE vs. V_{BE} (input voltage) for $V_{CE} = 1.00$ V for $V_{BE} = 0.4...1$ V. Plot also the I_B and give the current gain $\beta = I_{CE}/I_B$ vs. V_{BE} . (which is the so-called Gummel plot) (5 points)

3.40

Tasks 6 and 7: Formats to be used for the Answers in the PDF (Block 2 Part 1)

- (4) Give Table: S11 and S22 at 2.5 GHz for the model (blue) and the measured data (red). Compare the discrepancy (delta) in real and imaginary IMPEDANCE (i.e. in Ohms)! (5 points)
- (6) Give Table: S11 and S22: 2 sentences for magnitude and phase for model and data (5 points)
- (7) Give Table: S11 and S22: 2 sentences for magnitude and phase for model and data at 14 GHz (5 points)
- (8) Task: Give Table value S12 in absolute numbers at 14 GHz ? What quantity is the ratio of $|S_{21}|/|S_{12}|$? (3 sentences) (5 points)

3.41

Tasks 6 and 7: Formats to be used for the Answers in the pdf (Block 2 Part 1)

- (9) Plot Smith chart after 25 steps (5 points)
- (10) Plot Smith chart after 150 steps
- (11) Give 5 sentences max. (5 points)
- (12) Plot Smith chart (5 points)
- (13)Plot Magnitude and phase vs. f (5 points)
- (16) Plot curve tracer, 2 sentences (5 points)
- (17) Plot xy-plots with IC, IB and Beta (5 points)

3.42